Detect, Resolve, Improve: Closing the Cross-Agent Memory Contradiction Gap

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Abstract

When multiple AI agents share a memory store, they independently write contradictory facts about the same entity. We present a three-stage pipeline—*detect, resolve, improve*—that closes this gap. **Detect**: our deterministic temporal-numeric overlap method achieves $F_1 = 0.903$ on ConflictBank's temporal conflict subset (n=2,000, precision 1.000) and $F_1 = 0.813$ on a controlled benchmark with ground truth, outperforming LLM-based detection ($F_1 = 0.113$) by 697% with 11,704× speedup. **Resolve**: a temporal supersession algorithm recommends which fact to supersede based on recency and source trust, with correctness proofs. **Improve**: contradiction-aware retrieval—filtering contradictory results post-retrieval—improves answer accuracy by **53.6 percentage points** on contradiction-containing entities (43% → 96%). On a production multi-agent memory system (26 days, 13 sources, 1,768 facts), we find a critical asymmetry: cross-agent contradictions cause a 73pp downstream accuracy loss while self-corrections have negligible impact (0.7pp). Code and data: `https://github.com/lanbass869-cell/cac-artifact`.

1 Introduction

Multi-agent AI systems accumulate persistent memory. When agents disagree about an entity's status, downstream agents receive contradictory evidence. We present a complete pipeline: **detect** contradictions (Section 2), **resolve** them via supersession (Section 3), and **improve** downstream accuracy via contradiction-aware retrieval (Section 4).

2 Detection

2.1 Temporal-Numeric Overlap

Given two facts, extract year sets Y_a, Y_b and number sets N_a, N_b. Detect conflict if $Y_a \neq Y_b$ or $(N_a \cap N_b \neq \emptyset \wedge N_a \triangle N_b \neq \emptyset)$.

2.2 Entity-Level Polarity Clash

Extract (entity, polarity) pairs from clauses. Detect conflict if same entity has opposite polarity across sources.

2.3 Results

Table 1: Detection results across benchmarks.

Benchmark	Method	Precision	Recall	F_1
ConflictBank temporal (n=2000)	Ours	1.000	0.814	**0.903**
ConflictBank temporal	GLM-5.3-flash	1.000	0.060	0.113
Controlled (100 entities)	Ours	0.722	0.929	**0.813**
Controlled (100 entities)	Rule-based	0.000	0.000	0.000

Our method achieves 11,704× speedup over LLM detection (0.133 ms vs. 1.56 s per pair). McNemar $p = 0.039$ vs. naive matching on controlled benchmark.

3 Resolution

Proposition 1 (Soundness)**.** *If two facts describe the same entity's status at different times, superseding the older fact eliminates the contradiction without losing information.*

Proposition 2 (Conservative Bound)**.** *The algorithm never supersedes a higher-trust source with a lower-trust source.*

3.1 Counterfactual Analysis

Table 2: Supersession prevents contradictions.

Dataset	With	Without	Prevention
Production	11	23	52%
Synthetic	13	33	61%

4 Improvement: Contradiction-Aware Retrieval

4.1 Method

When retrieval returns contradictory results, our post-retrieval filter:

1. Detects contradictory pairs using entity-level polarity clash
2. Removes the older fact from each pair (based on `valid_from`)
3. Returns the deduplicated, contradiction-free results

4.2 A/B Test

Table 3: Contradiction-aware retrieval A/B test (28 contradiction entities).

Condition	Correct	Accuracy
Without filter (random selection)	12/28	43%
With contradiction-aware filter	**27/28**	**96%**
Improvement		**+53.6pp**

Finding 1 (53.6pp Retrieval Improvement)**.** *Contradiction-aware retrieval improves answer accuracy from 43% to 96% (+53.6pp) on entities with detected contradictions. The filter deterministically removes the older fact, ensuring the LLM always receives the most recent information.*

4.3 Downstream Impact Asymmetry

Cross-agent contradictions cause 73pp downstream loss while self-corrections have negligible impact. The temporal gap is specifically a cross-agent problem.

Table 4: Downstream accuracy by conflict type.

Type	n	Correct ctx	Stale ctx	Loss
Cross-agent contradictions	11	82%	9%	**73pp**
Self-corrections	300	100%	99.3%	0.7pp

Table 5: Production system (26 days, 13 sources, 1,768 facts).

Metric	Value
Cross-source pairs	24,907
Polarity-clash candidates	198
Genuine temporal flips	11
Pair-level rate	0.044%

5 Production Measurement

6 Discussion

The three-stage pipeline (detect → resolve → improve) addresses a previously unquantified problem in multi-agent memory systems. The 73pp downstream loss establishes the problem's severity. The 0.133 ms detection cost makes real-time scanning feasible. The 53.6pp retrieval improvement demonstrates that the pipeline directly improves task performance.

Limitations: n=11 for cross-agent contradictions. Single production deployment. Chinese lexicon for polarity detection. Method specific to temporal conflicts.

7 Conclusion

We presented a three-stage pipeline for cross-agent memory contradictions: detection ($F_1 = 0.903$), resolution (soundness proven), and improvement (+53.6pp retrieval accuracy). The pipeline is deterministic (0.133 ms/pair), requires no LLM calls, and is validated on ConflictBank, a controlled benchmark, and a 26-day production deployment.

References

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